Interview questions (with categories) related to the **Drupal-based Apigee Developer Portal** (often referred to as **Developer Portal (Drupal 8/9/10)** or **Apigee Kickstart**). These are suitable for roles involving **API management**, **developer experience**, **Drupal site building**, and **integration with Apigee Edge/Apigee X**.

🔹 General Questions: Drupal + Apigee Portal

1. What is the Apigee Developer Portal and what is its purpose?

Expected Answer:

- A developer portal is a **web interface** that exposes your APIs to external/internal developers.
- Built on **Drupal**, the Apigee Developer Portal allows devs to explore, register, and get access to APIs.
- Supports **API documentation, keys, rate plans, monetization**, etc.

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2. What is Apigee Kickstart?

Expected Answer:

- Apigee Kickstart is a **Drupal 9+ distribution** provided by Google Cloud to accelerate building a developer portal.
- Comes pre-configured with Apigee modules, styling, and best practices.
- Integrates easily with Apigee Edge and Apigee X.

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3. What are the main components/modules of the Apigee Developer Portal?

Expected:

- devconnect (older portals)
- apigee_edge (Drupal 8/9)
- apigee_api_product, apigee_consumer_app, apigee_register
- Integration with **SmartDocs** (for API documentation)

🔹 Integration & Configuration

4. How does the Apigee Developer Portal integrate with Apigee Edge or Apigee X?

- Via **API credentials (API key, secret)** stored in portal config.
- Uses Apigee Edge management APIs.

- Syncs API products, apps, developers, and proxies.

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5. How are API products managed in the Apigee portal?

- Synced from Apigee Edge into Drupal using the apigee_api_product module.
- API products can be displayed and selected when a developer registers an app.

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6. How do developers create and manage apps in the portal?

- Developers log in → Register → Create App → Select API Products.
- Upon app creation, a **consumer app** is created in Apigee, and an **API key** is generated.

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7. What is SmartDocs and how is it used in the Developer Portal?

- SmartDocs is a system for **interactive API documentation**.
- Allows developers to test APIs directly from the portal using Swagger/OAS specs.

🔹 Customization and Theming

8. How do you customize the Apigee Developer Portal UI?

- Use **Drupal theming layer** (.twig templates, preprocess hooks).
- Leverage Layout Builder or Panels for flexible page design.
- Override templates or use the apigee_subtheme provided by Kickstart.

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9. How can you extend the functionality of the Apigee portal?

- Create custom Drupal modules.
- Use Drupal's Form API and Hooks (e.g., hook_form_alter, hook_entity_presave).
- Integrate third-party systems (e.g., analytics, CRMs) via REST or GraphQL.

🔹 User Management & Security

10. How is user registration and login handled in the Apigee portal?

- Via Drupal's user system (user entity).
- May be extended with **OAuth, SSO (e.g., Okta)** using modules like openid_connect.

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11. How can access to APIs be controlled in the portal?

- API access is granted through **API Products**.
- Developer selects products during app registration.
- Permissions are controlled by roles and product access in Apigee Edge/X.

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12. How do you secure the developer portal and API access?

- HTTPS for all traffic.
- API keys/tokens generated securely.
- Drupal user permissions tightly controlled.
- Optional: Add CAPTCHA, rate limiting, reCAPTCHA, email verification.

🔹 **Advanced & Scenario-Based**

13. How would you integrate SSO (e.g., Okta) with the Developer Portal?

- Use openid_connect or simplesamlphp_auth Drupal modules.
- Map IdP attributes to Drupal user fields.
- Configure role mapping for access control.

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14. How would you migrate from the classic (Drupal 7) dev portal to Apigee Kickstart (Drupal 9/10)?

- Audit existing D7 site for content and custom modules.
- Use migrate module to bring over users, content types, API products.
- Rebuild theming and custom modules in D9.
- Reconnect with Apigee via new credentials.

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15. What are some common challenges working with Apigee Developer Portals?

Possible answers:

- Keeping portal data in sync with Apigee Edge.
- Complex theming for enterprise design standards.
- Managing user roles and access securely.
- Integrating 3rd-party systems like CRM, analytics, SSO.

📘 Bonus: Apigee Portal DevOps / Deployment Questions

16. How do you deploy the Apigee Developer Portal in production?

- Use Composer-based workflows (especially for Kickstart).
- Use tools like Acquia, Pantheon, or Kubernetes.
- Store API credentials in **secrets** or **environment config**.
- Ensure CI/CD pipeline runs DB updates, config import, and cache clear.

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17. What is the role of Composer in the Apigee portal setup?

- Composer manages dependencies (Drupal core, modules, libraries).
- Kickstart provides a base composer.json to start quickly.

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18. What are monetization features in the Apigee Developer Portal?

- If enabled in Apigee, developers can select **rate plans**, track **usage**, and access **billing features**.
- Requires Apigee Monetization to be configured.

✅ Summary

Category	Focus
Basics	Purpose, components, Kickstart
Integration	Apigee Edge/X, API Products, SmartDocs
Customization	Themes, Layout Builder, custom modules
Security	Auth, SSO, user access
DevOps	Composer, CI/CD, hosting

Would you like a **PDF version**, or should I prep **mock interview answers** or **real-time practice questions**?

Integrate the Drupal portal with the Apigee platform using standard and custom Apigee modules

Integrating a **Drupal portal** with the **Apigee platform** is a common approach for API-driven digital experiences, especially in developer portals. This integration allows developers to discover, register, and consume APIs managed through Apigee, all within a branded Drupal interface.

Here's a step-by-step guide for integrating **Drupal** (usually Drupal 9 or 10) with **Apigee**, using both **standard modules** and **custom implementations** when necessary.

🧩 Overview of Integration

What You'll Achieve:

- Display and manage APIs published via Apigee in Drupal.
- Allow developers to register and get API keys.
- Enable API product subscription, usage tracking, and documentation (SmartDocs).

✅ Prerequisites

1. **Drupal site installed** (preferably Drupal 9 or 10).
2. **Apigee account** (Edge or X, SaaS or hybrid).
3. **Apigee Developer Portal credentials** (API Client ID/Secret).
4. Admin access to Drupal and Apigee.

🔌 Step 1: Install the Apigee Modules

Standard Drupal Modules from Apigee:

Install the following contributed modules (most are on Drupal.org):

1. Apigee Edge
2. Apigee Edge Auth
3. Apigee Edge Developer
4. Apigee Edge API Product
5. Apigee SmartDocs

You can install via Composer:

```
composer require drupal/apigee_edge drupal/apigee_edge_auth drupal/apigee_edge_developer drupal/apigee_edge_api_product drupal/smartdocs
```

Then enable them:

```
drush en apigee_edge apigee_edge_auth apigee_edge_developer apigee_edge_api_product smartdocs -y
```

🔑 Step 2: Configure Apigee Edge Connection

1. Go to **Configuration > Apigee Edge > Settings**.
2. Fill in:
   - **Organization name**
   - **Apigee Environment**
   - **Apigee Developer Portal credentials**
   - Select **Authentication Method** (basic or OAuth).
3. Save configuration.

🧑‍💻 Step 3: Enable Developer Workflows

1. Use the **Apigee Edge Developer** module to allow users to:
   - Register as developers.
   - Manage their apps.
   - Get API keys.
2. Use **Views** and **Permissions** to expose the appropriate UI for users to:
   - View their subscriptions.
   - Create and manage apps.

📄 Step 4: Import and Display API Products

1. Use the **Apigee Edge API Product** module to import your products from Apigee.
2. Products will be created as Drupal entities.
3. You can customize display using Views or custom templates.

📘 Step 5: Enable SmartDocs (Interactive API Docs)

1. Install and enable the **SmartDocs** module.
2. Fetch API specs (Swagger/OpenAPI) from Apigee Edge.
3. Render interactive API docs inside Drupal.
   This is particularly useful for public APIs, allowing developers to test endpoints directly from the portal.

⚙️ Step 6: Custom Modules (Optional)

If your use case is not entirely met by the standard modules, you can create custom modules for:

Examples of Customizations:

- Custom workflows for app approval or API subscription.
- Integration with third-party authentication (e.g., SSO).
- Custom reporting dashboards using Apigee analytics APIs.
- Granular permissions or roles for managing APIs or developers.
- Custom API search interface.

Example: Custom module to fetch Apigee analytics

```
$response = \Drupal::httpClient()->get('https://api.enterprise.apigee.com/v1/organizations/{org}/environments/{env}/stats');
$data = json_decode($response->getBody(), true);
```

🔒 Step 7: Security and Access Control

- Use Drupal roles/permissions to limit access to developer features.
- Apply rate limiting or additional security on forms (e.g., API key generation).
- Enable HTTPS and secure cookies.

🧪 Step 8: Test the Integration

Make sure to test:

- Developer registration and login.
- API product discovery.
- App creation and API key retrieval.
- SmartDocs rendering and testing.
- Analytics/dashboard features.

🚀 Bonus: Apigee Monetization Support

If you use **Apigee Monetization**, you'll likely need custom modules to:

- Display pricing plans.
- Enable paid API subscriptions.
- Show usage reports and invoices.

This is usually not handled fully by standard modules and will require API integrations with Apigee's monetization endpoints.

🔁 Maintenance and Syncing

- Use Drupal cron or scheduled jobs to sync APIs/products if managed externally.
- Monitor Apigee API rate limits and quotas.

🎯 Final Notes

Feature	Standard Module	Customization Needed?
API Product Import	✅	❌
Developer Registration	✅	Maybe (for custom UX)
App Management / API Keys	✅	❌
SmartDocs (API Docs)	✅	❌
Analytics	❌	✅
Monetization	❌	✅
SSO or Custom Auth	❌	✅

Utilize Apigee's API Catalog and SmartDocs to automatically publish API documentation within the portal.

To **automatically publish API documentation within a Drupal portal** using **Apigee's API Catalog** and **SmartDocs**, you'll combine Apigee's discovery capabilities with Drupal's rendering and content management tools. The goal is to:

> Automatically retrieve APIs and documentation from Apigee and render them as SmartDocs pages in Drupal.

🎯 What You'll Achieve

- Auto-import API specifications (OpenAPI/Swagger) from Apigee's **API Catalog**.
- Render those specifications in **SmartDocs**.
- Keep the documentation up to date with Apigee.

🧱 Components Involved

Component	Purpose
Apigee API Catalog	Stores and manages OpenAPI specs
Apigee SmartDocs module (Drupal)	Renders interactive documentation
Apigee Edge modules	Handles API product sync, developer workflows
Custom automation (optional)	For periodic fetching/publishing

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🧩 Step-by-Step Integration

🔧 Step 1: Set Up the Required Modules

Install Apigee-related Drupal modules (if not already done):

```
composer require drupal/apigee_edge drupal/smartdocs
drush en apigee_edge apigee_edge_api_product smartdocs -y
```

🔐 Step 2: Configure Connection to Apigee

In **Drupal Admin UI**:

Path:
/admin/config/services/apigee-edge/settings

Enter:

- Organization
- Environment
- Apigee credentials

- Authentication method (OAuth preferred)

Save settings.

📥 Step 3: Fetch API Specs from Apigee API Catalog

Apigee stores OpenAPI documents in its **API Catalog**. To fetch them:

Apigee API Endpoint (Example):

```
GET /v1/organizations/{org}/apicatalog/documents
Authorization: Bearer {access_token}
```

Each document typically includes:

- API name
- Version
- Spec in JSON or YAML

You can test this with tools like Postman or curl to confirm connectivity.

📦 Step 4: Import into SmartDocs

There are two main approaches:

✅ Option A: Use UI to Import

1. Go to **SmartDocs > Add API Specification**:
   `/admin/config/services/smartdocs/api`
2. Paste or upload the OpenAPI JSON/YAML spec.
3. SmartDocs will:
   - Parse the spec.
   - Generate Drupal entities for endpoints.
   - Create rendered docs with testing console.

🌀 Option B: Automate with Custom Module

Create a **custom module** to:

- Call Apigee API Catalog
- Retrieve OpenAPI spec
- Import into SmartDocs programmatically

Sample PHP Snippet:

```
use Drupal\smartdocs\Entity\SmartDocsAPI;

// 1. Fetch OpenAPI spec from Apigee
$spec_url = 'https://apigee.googleapis.com/v1/organizations/{org}/apicatalog/documents/{doc_id}';
$response = \Drupal::httpClient()->get($spec_url, [
  'headers' => [
```

```
    'Authorization' => 'Bearer ' . $access_token,
  ],
]);
$openapi_spec = $response->getBody()->getContents();

// 2. Create SmartDocs API entity
$api_entity = SmartDocsAPI::create([
  'label' => 'My API',
  'spec' => $openapi_spec,
  'format' => 'json', // or 'yaml'
]);
$api_entity->save();
```

Schedule this via cron to **keep API docs in sync**.

🧪 Step 5: Test Documentation Rendering

1. Navigate to the API SmartDocs page in Drupal.
2. You should see:
    - Grouped API endpoints
    - Request/response schemas
    - Try-it-out console (authenticated with API key)

You can control layout using Views or Blocks.

🔁 Step 6: Automate Syncing (Optional)

Use Drupal's hook_cron() or **Scheduled Tasks** to automate:

- Pulling API specs from Apigee
- Comparing hashes or version
- Updating SmartDocs only if changed

📘 Additional Enhancements

Feature	Approach
Group APIs by category	Use Drupal Taxonomy
Add version control	Use separate SmartDocs entities per version
Enable Try-it-out	Requires API key integration
Private/Public APIs	Use Drupal roles/permissions
Customize SmartDocs styling	Override templates and CSS

📌 Notes and Best Practices

- SmartDocs currently supports **OpenAPI 2.0 (Swagger)** best; limited support for 3.x.

- Use **access tokens** securely—store them in environment variables or Drupal's key management.
- For large API catalogs, consider paginating or batching imports.
- Implement **rate limiting** when fetching large data sets from Apigee.

✅ Summary

Step	Action
1	Install Apigee + SmartDocs modules
2	Configure Apigee connection
3	Retrieve OpenAPI specs from API Catalog
4	Import specs into SmartDocs (manually or via custom module)
5	Render and test API docs
6	Automate sync (optional but recommended)

Would you like a **ready-to-use Drupal custom module scaffold** that automates fetching and importing SmartDocs from Apigee? I can generate one.

Configure and manage API key registration, user synchronization, and access control between Apigee and Drupal.

To **configure and manage API key registration, user synchronization, and access control** between **Apigee** and a **Drupal developer portal**, you'll need to leverage both standard Apigee modules and some customizations to ensure seamless interoperability between the systems.

This setup enables developers to:

- Register in Drupal
- Get synchronized as **developers in Apigee**
- Create **apps** in Drupal
- Automatically get **API keys (client ID/secret)**
- Enforce access control on API product subscriptions

✅ Key Goals

Feature	Outcome
🔐 User Management	Drupal users are synced with Apigee developers
🔑 API Key Registration	Developers can create apps and get client credentials
🚾 Access Control	API product visibility and subscriptions are permission-based

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📦 Required Modules

Install these modules if not already set up:

```
composer require drupal/apigee_edge drupal/apigee_edge_auth drupal/apigee_edge_developer drupal/apigee_edge_api_product
drush en apigee_edge apigee_edge_auth apigee_edge_developer apigee_edge_api_product -y
```

🔧 Step-by-Step Configuration

🧑‍💻 1. User Registration and Synchronization

✅ Enable developer sync:

- Navigate to:
  Admin → Configuration → Services → Apigee Edge → Developer Settings

✅ Settings to configure:

- **Auto-create Apigee developer on user registration**
- **Map Drupal fields to Apigee developer fields** (e.g., name, email)

This ensures that when a user signs up in Drupal, a **matching developer entity** is created in Apigee.

🔁 **Bi-directional Sync (optional):**

To allow updates from Apigee → Drupal (e.g., user status), you may need custom logic or periodic sync via a custom module or cron job.

🔑 2. API Key (App) Registration

Once a user is registered:

✅ **Allow them to create apps:**

- Enable the **App Creation Form** route:
  - /developer/apps/add

This form allows users to:

- Create a **developer app**
- Select **API Products**
- Receive the **API key (client_id / secret)** automatically

Apigee handles API key generation. Drupal exposes it via the apigee_edge_developer_app entity.

🔁 3. Sync Apps Between Drupal and Apigee

When a developer creates or edits an app in Drupal:

- It automatically creates or updates the corresponding app in Apigee
- Subscribes to selected API products
- Generates API keys

✅ You can view apps:

- Admin UI: /admin/content/developer-apps
- User account: /developer/apps

🔐 4. Access Control & Permissions

Use Drupal's **role-based permissions** system:

Suggested roles:

- anonymous
- authenticated
- developer
- api_admin

Set permissions at:

/admin/people/permissions

Key permissions to assign:

- **Create/edit developer apps**
- **Subscribe to API products**
- **View API keys**
- **Access SmartDocs**

Example:

Only users with the developer role can:

- View API products
- Register apps
- Access documentation with test console

📦 5. Control API Product Visibility

API Products are imported as **Drupal entities**:

- Entity type: apigee_edge_api_product
- Path: /admin/content/api-products

You can:

- Add fields (e.g., taxonomy, visibility flags)
- Create Views to show/hide products
- Use permissions or field-based access control

Example:

```
// Only show public products:
$query = \Drupal::entityTypeManager()
  ->getStorage('apigee_edge_api_product')
  ->getQuery()
  ->condition('field_visibility', 'public');
```

🔐 6. Secure Access to API Keys

By default, developers can see their own API keys. You can:

- Mask API keys in templates
- Limit edit access via field-level permissions
- Log API key downloads for auditing

🧪 7. Test the Flow End-to-End

Test Scenario:

Action	Outcome
Register as new user	→ Developer created in Apigee
Log in as developer	→ View available API products

Create an app	→ API key generated in Apigee
Subscribe to product	→ App linked to product in Apigee
Use key in SmartDocs	→ Endpoint test authorized

🛠️ 8. Optional Customizations

🔄 Cron-Based Sync:

Use hook_cron() or custom scheduled tasks to:

- Sync developer status
- Update API key status (revoked, expired)
- Pull new API product changes from Apigee

📈 Analytics Integration:

Use Apigee's analytics APIs to show usage dashboards in Drupal:

GET /v1/organizations/{org}/environments/{env}/stats/api

Display in a custom dashboard view or block for each developer.

🔁 Summary Workflow Diagram

[User Registers in Drupal]
↓
[Developer Entity Created in Apigee]
↓
[User Logs In → Sees API Products]
↓
[User Creates App → API Key Generated in Apigee]
↓
[User Uses Key via SmartDocs or External Clients]

✅ Final Recommendations

Task	Module	Custom Needed?
User registration & sync	apigee_edge_developer	❌
App and API key generation	apigee_edge_developer_app	❌
API product management	apigee_edge_api_product	❌
Product visibility rules	Drupal Views or field conditions	✅
Custom workflows (e.g., approvals)	Custom module	✅
Developer analytics dashboard	Custom integration with Apigee API	✅

Build custom features that leverage Apigee APIs, such as displaying API analytics or managing team permissions

Building **custom features in Drupal** that leverage **Apigee's APIs** is a powerful way to extend your developer portal beyond what's possible with the out-of-the-box modules.

Here are **two common use cases** you requested:

✅ **Custom Feature 1: Display API Analytics**

🔍 **Use Case:**

Show **API usage statistics** (calls, errors, latency) to:

- Individual developers (for their apps)
- Admins (for all apps/products)

🔧 **How It Works**

Use Apigee's **Management API** to fetch analytics data, then render it in Drupal using:

- Custom modules
- Blocks, Views, or Dashboards

🧪 **Example 1: Developer App Analytics**

🧠 **Apigee Endpoint:**
GET /v1/organizations/{org}/environments/{env}/apps/{app_id}/stats

- Parameters: select=sum(message_count),avg(response_time)
- Time range: &timeRange=30d

🛠️ **Drupal Custom Module Snippet**

File: apigee_analytics/src/Controller/AnalyticsController.php

```
namespace Drupal\apigee_analytics\Controller;

use Drupal\Core\Controller\ControllerBase;
use Symfony\Component\DependencyInjection\ContainerInterface;
use GuzzleHttp\ClientInterface;

class AnalyticsController extends ControllerBase {

  protected $httpClient;

  public function __construct(ClientInterface $http_client) {
    $this->httpClient = $http_client;
```

```
  }

  public static function create(ContainerInterface $container) {
    return new static(
      $container->get('http_client')
    );
  }

  public function getAppAnalytics($app_id) {
    $org = 'your-org';
    $env = 'test'; // or prod
    $token = 'your-apigee-access-token'; // securely load from config or key vault

    $response =
$this->httpClient->get("https://apigee.googleapis.com/v1/organizations/{$org}/environments/{$env}
/apps/{$app_id}/stats?select=sum(message_count),avg(response_time)&timeRange=30d", [
      'headers' => [
        'Authorization' => "Bearer {$token}",
      ],
    ]);

    $data = json_decode($response->getBody(), true);

    return [
      '#theme' => 'apigee_app_analytics',
      '#data' => $data,
    ];
  }
}
```

✅ **Output as Block or Dashboard:**

Use a custom Twig template or create a block plugin to render the stats with charts (e.g., Chart.js).

✅ Custom Feature 2: Manage Team Permissions

🧍 **Use Case:**

Allow multiple developers to collaborate on apps in Apigee via **teams** or **shared app ownership**.

🔧 **How It Works**

Apigee doesn't support teams natively, but you can implement it in Drupal using:

- A custom **"Team" entity** (Content type or Config entity)
- Relationships to **Drupal users** and **Apigee apps**
- Permission control via Drupal roles

🧱 Implementation Strategy

1. **Create Custom Entity: team**
   - Fields: name, members (user references), associated apps, roles (manager/developer)
2. **Assign Apps to Teams**
   - Extend developer app form to include a "Team" field
   - Only show/edit if user is part of the team
3. **Custom Access Checks**
   - Use hook_entity_access() or an access control handler to restrict app editing/viewing

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🧩 Entity Definition Snippet

apigee_team/src/Entity/Team.php (simplified)

```
use Drupal\Core\Entity\ContentEntityBase;

/**
 * @ContentEntityType(
 *   id = "apigee_team",
 *   label = @Translation("Apigee Team"),
 *   base_table = "apigee_team",
 *   handlers = {
 *     "view_builder" = "Drupal\Core\Entity\EntityViewBuilder",
 *     "access" = "Drupal\apigee_team\TeamAccessControlHandler",
 *     ...
 *   },
 *   ...
 * )
 */
class Team extends ContentEntityBase {
  // define fields like team name, members, app refs
}
```

🔐 Access Control Handler

TeamAccessControlHandler.php

```
class TeamAccessControlHandler extends EntityAccessControlHandler {
  protected function checkAccess(EntityInterface $entity, $operation, AccountInterface $account) {
    // Only allow team members to view or edit
    $members = $entity->get('field_team_members')->referencedEntities();
    foreach ($members as $member) {
      if ($member->id() == $account->id()) {
        return AccessResult::allowed();
      }
    }
    return AccessResult::forbidden();
  }
```

```
}
```

🔁 Bonus: Sync Apigee App Ownership with Drupal Teams

- When a user creates an app in Drupal, store the **Team ID**.
- Store team in Apigee as a **custom attribute** on the app:

```
"attributes": [
  {
    "name": "team_id",
    "value": "team_123"
  }
]
```

Use Apigee API:

```
POST /v1/organizations/{org}/developers/{email}/apps/{app}/attributes
```

🧪 Admin View of Team Apps

Use Drupal Views to:

- List all apps per team
- Show team members and their roles
- Filter analytics by team

🎯 Summary Table

Feature	Approach	Module Needed
API usage stats	Apigee Analytics API + Custom Controller	✅
Display charts	Chart.js or Views integration	Optional
Developer teams	Custom entity + field mapping	✅
App access control	Access handlers / entity policies	✅
Sync app metadata	Apigee App attributes API	✅

✅ Ready-to-Use Module Offer

Would you like me to generate a **ready-to-use custom Drupal module** for:

- 📊 Analytics dashboard
- 👥 Team management and app sharing
- 🔐 App-level access control

Plan and execute upgrades of the Drupal core, contributed modules, and Apigee-specific modules to ensure security, performance, and access to new features.

Upgrading the **Drupal core**, **contributed modules**, and **Apigee-specific modules** is crucial for maintaining **security**, **performance**, and **feature parity**. Below is a detailed **plan** and **execution strategy** you can follow:

🔧 Phase 1: Planning and Assessment

1. Inventory Audit

- List all currently installed:
  - Drupal core version.
  - Contributed modules.
  - Custom or Apigee-specific modules.
  - Themes (especially if they're custom or have dependencies).
- Tools: drush pm:list, composer show, or manually from the admin UI.

2. Identify Available Updates

- Use:
  - drush pm:security (security updates).
  - drush pm:updates (all updates).
  - composer outdated (composer-managed sites).
- Check Drupal.org for release notes and compatibility information.

3. Check Compatibility

- Ensure compatibility of contributed and Apigee modules with the target Drupal core version.
- Visit the project page on Drupal.org or vendor documentation for Apigee modules.

4. Backup Everything

- Full backup of:
  - Database.
  - Codebase.
  - Files directory.
- Tools: drush sql-dump, rsync, tar, or your hosting provider's backup tools.

🏗️ Phase 2: Prepare and Test in Staging

5. Clone to Staging

- Use a separate environment or a local dev setup (Docker, DDEV, Lando, etc.).
- Validate that all features work as expected before upgrades.

6. Update Core & Modules

a. Drupal Core:

```
composer update drupal/core drupal/core-* --with-all-dependencies
```

b. Contributed Modules:

```
composer update drupal/module_name --with-dependencies
```

c. Apigee-specific Modules:

- Example: apigee_edge, devportal modules.
- If installed via Composer:

```
composer update drupal/apigee_edge drupal/devportal --with-dependencies
```

- Review Apigee documentation for compatibility and patch notes.

7. Run Database Updates

```
drush updatedb
```

8. Clear Cache and Rebuild

```
drush cr
```

9. Regression Testing

- Automated and manual QA tests:
  - APIs accessible via Dev Portal.
  - Module functionality.
  - Content creation/editing.
  - Custom forms/views.
- Use tools like:
  - Behat or Cypress for automated tests.
  - BrowserStack for cross-browser testing.

🚀 Phase 3: Deployment to Production

10. Prepare Production

- Schedule a maintenance window if necessary.
- Put site in maintenance mode:

```
drush sset system.maintenance_mode 1
```

```
drush cr
```

11. Deploy Code and Run Updates

- Deploy updated code via Git or CI/CD.
- Run:

```
drush updatedb
```

```
drush cr
```

12. Post-deployment QA

- Spot-check critical paths.
- Monitor logs:

```
drush ws
```

13. Turn off Maintenance Mode

```
drush sset system.maintenance_mode 0
```

```
drush cr
```

📈 Phase 4: Ongoing Maintenance

- Set up monitoring for:
  - Security advisories from Drupal.
  - Apigee module updates.
- Automate checks with `drush pm:security` in cron or CI.
- Keep a changelog of updates with release notes for auditing.

🧩 Tips for Apigee-specific Modules

- Apigee modules often interact with:
  - API Products
  - Developer apps
  - Apigee Edge/Hybrid
- Always test:
  - OAuth flows
  - API publishing workflows
  - Apigee Edge connectivity post-upgrade

Would you like a tailored checklist or automated script (Bash or Ansible) to assist in this process?

Manage patches, security updates, and bug fixes for the Drupal environment

Managing **patches**, **security updates**, and **bug fixes** in a **Drupal environment** is critical to maintaining site integrity, performance, and security compliance. Below is a structured approach to **effectively manage and automate** this ongoing process.

🔐 Goal

Ensure the Drupal environment (core, contributed modules, themes, custom code) is:

- Secure (patched against known vulnerabilities)
- Up-to-date (with fixes and improvements)
- Stable (free of regressions or performance issues)

✅ 1. Establish Update & Patch Management Policies

Define:

- **Update frequency**: (e.g., weekly checks, monthly minor updates, immediate critical patches)
- **Patch approval process**: Involve dev leads, QA, or security officers
- **Environments for testing**: Dev → Stage → Prod

🛠️ 2. Tools and Automation Setup

a. Use Composer for Dependency Management

- Centralized control of Drupal core and modules
- Enables patch management with cweagans/composer-patches

composer require cweagans/composer-patches

b. Use Drush for CLI Updates & Security Checks

drush pm:security # List security vulnerabilities

drush pm:updates # Show available updates

drush updatedb # Apply database updates

c. CI/CD Pipelines

- Integrate tools like GitHub Actions, GitLab CI, or Bitbucket Pipelines to:
  - Run security checks
  - Apply updates on branches
  - Notify or auto-create PRs for updates

🔁 3. Security Updates Workflow

Weekly Maintenance Routine:

Check for updates:

```
composer outdated "drupal/*"
```

```
drush pm:security
```

1. **Update secure modules**:

   ```
   composer update drupal/[module_name] --with-dependencies
   ```
2. **Apply database updates**:

   ```
   drush updatedb
   ```
3. **Clear cache**:

   ```
   drush cr
   ```
4. **Critical Patch Workflow (ASAP):**
1. Monitor:
   - Drupal Security Advisories
   - Automated alerts from security mailing list or RSS
2. Patch immediately in dev → stage → prod
3. Perform regression testing (esp. for critical paths and integrations)
4. Document actions in change log / security log

🧩 4. Patch Management for Core & Modules

Use Composer Patches

Add to composer.json:

```
"extra": {
  "patches": {
    "drupal/module_name": {
      "Patch title": "https://www.drupal.org/files/issues/issue-id-description.patch"
    }
  }
}
```

Then run:

```
composer install
```

> ✅ Always note the Drupal.org issue URL in comments for traceability.

🔍 5. Bug Fix Management

a. Triage

- Use an internal issue tracker (Jira, GitHub, etc.)
- Label bugs: severity, module, regression, etc.

b. Fix Workflow

1. Create branch or patch
2. Unit and functional test
3. Code review + merge
4. Deploy to staging → regression test → deploy to production

---

🔄 6. Custom Module & Theme Updates

- Maintain changelogs in /custom/[module]/CHANGELOG.md
- Follow Drupal API changes and deprecations
- Regularly audit custom code:
  - Use Upgrade Status
  - Use PHPStan or Drupal Check for deprecated API usage

🧪 7. Testing Strategy

- **Automated Tests**: PHPUnit, Behat, Cypress for regressions
- **Manual Smoke Testing** after updates
- Integration testing for Apigee and APIs
- Visual regression tools (e.g. BackstopJS)

📝 8. Logging & Documentation

- Maintain an update log: module/core name, version, reason (security/bugfix), date
- Document patch history (e.g., in Git commits or README in /patches/ folder)
- Track applied patches in source control

📅 Example Maintenance Schedule

Task	Frequency	Responsible
Security check (drush)	Weekly	DevOps / Developer
Composer outdated check	Weekly	Developer
Critical patching	ASAP	Security / Dev
Full module/core updates	Monthly	Developer
Custom code review	Quarterly	Architect / Lead
Drupal core upgrade (minor)	Quarterly	Developer

Would you like a sample automation script (Bash/CI) or a template for tracking patches and updates?

Great! Below are **two resources** to help automate and track your **Drupal patch and update process**:

✅ 1. Sample Bash Script – Drupal Update & Patch Automation

This script helps automate checking for updates, applying them, and managing patches for a Composer-managed Drupal project.

🔧 drupal_update.sh

```
#!/bin/bash

set -e

echo "🔍 Checking for security updates..."

# Run Drush security check

drush pm:security

echo "📦 Checking for outdated packages..."

composer outdated "drupal/*"

echo "🔁 Updating Drupal core and contributed modules..."

composer update drupal/core drupal/core-* "drupal/*" --with-all-dependencies

echo "🩹 Reapplying any defined patches..."

composer install

echo "📂 Running database updates..."

drush updatedb -y

echo "🧹 Clearing cache..."

drush cr

echo "✅ Update process complete!"
```

🔧 Usage

```
chmod +x drupal_update.sh

./drupal_update.sh
```

> 💡 You can run this via a cron job or CI pipeline for weekly maintenance.

📋 2. Patch & Update Tracking Template (Markdown)

Use this to **log and document** all module updates, patches applied, and related notes. Store it in a Git-tracked location like docs/patch-log.md.

📄 docs/patch-log.md

🛠️ Drupal Patches & Update Log

🔄 Core Updates

Date	Version	Reason	Notes
2025-09-21	10.3.2	Security	SA-CORE-2025-005 applied

📦 Module Updates

Date	Module	Version	Type	Reason	Notes
2025-09-20	Webform	6.0.3	Contrib	Bugfix	Fixed form validation bug
2025-09-20	Apigee Edge DevPortal	2.23	Apigee	Compatibility	Supports Drupal 10.3

🩹 Applied Patches

Date	Module	Patch Name	Source (URL)	Notes
2025-09-19	apigee_edge	Token issue patch	https://www.drupal.org/files/issues/xyz.patch	Temporary fix from issue
2025-09-15	menu_link	Menu weight patch	Local: patches/menu-link-fix.patch	Internal bug

🔐 Security Events

Date	Advisory ID	Affected Module/Core	Action Taken	Notes
2025-09-21	SA-CORE-2025-005	Core 10.3.1	Upgraded to 10.3.2	No regressions found

---

➕ Optional: Composer Patch Sample

In composer.json:

```
"extra": {
  "patches": {
    "drupal/apigee_edge": {
      "Fix API token expiration bug": "https://www.drupal.org/files/issues/1234567-apigee-token-expiry.patch"
    }
  }
}
```

🚀 Want to Integrate into CI/CD?

I can also provide:

- A GitHub Actions workflow
- GitLab CI .gitlab-ci.yml
- Bitbucket Pipelines configuration

Let me know your CI system, and I'll tailor it.

o Implement and maintain CI/CD pipelines for portal deployments and updates.

Implementing and maintaining **CI/CD pipelines using Jenkins** for a **Drupal-based developer portal (e.g., Apigee DevPortal)** involves automating the process of:

- **Building** the Drupal codebase (with Composer),
- **Validating** code and configs (lint, security scan),
- **Deploying** to remote servers (dev/stage/prod),
- **Running updates** (database, config sync, cache),
- Optionally **notifying** teams (Slack, email).

🚀 CI/CD with Jenkins for Drupal Portals – Overview

🔧 Requirements

- Jenkins installed (preferably with Blue Ocean or Pipeline plugin)
- SSH access to deployment targets (dev/stage/prod)
- Drupal site managed via **Composer**
- Jenkins agent with PHP, Composer, Drush, Git, SSH

🧱 Jenkins Pipeline Structure

Stages:

1. **Checkout Code**
2. **Install Dependencies**
3. **Lint / Audit**
4. **Optional: Run Tests**
5. **Deploy to Serve**
6. **Post-deployment Tasks**
7. **Notify / Archive Logs**

🗂️ Example Jenkins Pipeline (Declarative)

📄 **Jenkinsfile**

```
pipeline {
    agent any
    environment {
        DEPLOY_USER = 'deploy'
        DEPLOY_HOST = 'dev.example.com'
        DEPLOY_PATH = '/var/www/portal'
        SSH_KEY_CREDENTIALS_ID = 'ssh-deploy-key' // Jenkins SSH credential ID
    }
    stages {
        stage('🧾 Checkout') {
            steps {
                checkout scm
            }
        }
        stage('⚙️ Install Dependencies') {
```

```
        steps {
            sh 'composer install --no-interaction --prefer-dist'
        }
    }
    stage('🧪 Lint & Security Checks') {
        steps {
            sh 'composer validate'
            sh 'composer audit'
            sh 'vendor/bin/drush pm:security'
        }
    }
    stage('🚀 Deploy to Server') {
        steps {
            sshagent([env.SSH_KEY_CREDENTIALS_ID]) {
                sh """
                    rsync -az --exclude=.git --exclude=vendor/ ./
${DEPLOY_USER}@${DEPLOY_HOST}:${DEPLOY_PATH}
                    ssh ${DEPLOY_USER}@${DEPLOY_HOST} << EOF
                      cd ${DEPLOY_PATH}
                      composer install --no-dev --optimize-autoloader
                      vendor/bin/drush updatedb -y
                      vendor/bin/drush cr
                      vendor/bin/drush cim -y
                    EOF
                """
            }
        }
    }
    stage('✅ Post-deploy Check') {
        steps {
            sshagent([env.SSH_KEY_CREDENTIALS_ID]) {
                sh """
                    ssh ${DEPLOY_USER}@${DEPLOY_HOST} "cd ${DEPLOY_PATH} &&
vendor/bin/drush status"
                """
            }
        }
    }
  }
  post {
    success {
        echo '✅ Deployment Successful'
    }
    failure {
        echo '❌ Deployment Failed'
    }
  }
}
```

🔐 Jenkins Configuration

1. Add SSH Credentials

- Go to: **Jenkins > Manage Jenkins > Credentials**
- Type: **SSH Username with private key**
- ID: ssh-deploy-key (as referenced in the Jenkinsfile)

2. Install Required Plugins

- **Pipeline**
- **SSH Agent**
- **Credentials**
- **Blue Ocean** (optional, UI/UX)
- **AnsiColor** (optional, for colorful console output)

3. Agent Requirements

Ensure the Jenkins agent (or controller if used directly) has:

```
php -v
composer -V
drush --version
git --version
```

🧪 Optional Add-ons

Feature	Tool/Integration	Notes
Automated Tests	PHPUnit, Behat	Add test stage in pipeline
Slack Notification	Slack Webhook Plugin	Notify on success/failure
Multi-env Deploy	Parameterized Pipeline	Deploy to dev/stage/prod
Blue-Green Deploy	Custom Scripts	For high availability

📦 Directory & Project Best Practices

```
project-root/
├── composer.json
├── drush/
│   └── drush.yml
├── patches/
├── web/
│   └── index.php
├── Jenkinsfile
└── .env
```

🛡️ Maintenance Tasks

Task	Frequency	Jenkins Setup
Composer Audit	Weekly	Schedule job
Drush Security Checks	Weekly	CLI + Email report
Deploy to Dev	On push	CI trigger
Deploy to Prod	Manual	Protected step
Database Backup	Before deploy	Custom SSH script

🧩 Apigee DevPortal Considerations

- Post-deploy, validate:
  - API product sync via drush apigee-edge:status
  - Developer login / OAuth flows
  - Portal customizations & themes
- You can add specific **smoke tests** or curl calls in the Jenkins pipeline to verify API docs or dev sign-up flows.